

RESEARCH PROPOSAL

Development of a Machine Learning-Based Surface Water Quality Prediction Model for the Sibalom River, Leon, Iloilo Using Regional Water Quality Data

1. Introduction and Rationale

The deterioration of river water quality poses significant threats to both aquatic ecosystems and human health. In the province of Iloilo, effective water quality monitoring is essential for agricultural, industrial, and domestic resource management. While comprehensive physicochemical assessments have been conducted on major water bodies such as the Jalaur River System—which spans 123 kilometers and serves as a primary source of irrigation and potable water [2]—there remains a critical data sparsity challenge for the Sibalom River in Leon, Iloilo.

Traditional water quality monitoring relies heavily on time-consuming laboratory testing of parameters such as Biological Oxygen Demand (BOD), Total Coliform Bacteria (TCB), and Dissolved Oxygen (DO). To address the lack of continuous sampling in the Sibalom River, this study proposes a machine learning (ML) framework that utilizes transfer learning. By training a robust computational model on the extensive 2020-2025 water quality dataset from the Jalaur River System Water Quality Management Area (JRS-WQMA) [1], the model can infer and predict the ecological suitability and Water Quality Index (WQI) of the Sibalom River based on limited input parameters.

2. Objectives of the Study

General Objective: To develop, optimize, and evaluate a machine learning-based classification and prediction model capable of estimating the Water Quality Index (WQI) of the Sibalom River by transferring learned relationships from JRS-WQMA regional data.

Specific Objectives:

- To compute the baseline Water Quality Index (WQI) for the 2020-2025 JRS-WQMA historical dataset using the Weighted Arithmetic Water Quality Index method [4].
- To perform rigorous data preprocessing using K-Nearest Neighbors (KNN) for missing value imputation and Random Forest (RF) for outlier detection [3].

- To isolate the most critical physicochemical predictors (e.g., pH, DO, Temperature) by applying a comprehensive weighting method that combines Information Entropy and Pearson Correlation Coefficients [5].
- To design and train Python-based ML classifiers (XGBoost, Support Vector Machines, Multilayer Perceptron) to categorize water samples into DENR-standardized suitability classes [3, 4].
- To evaluate the performance of the algorithms using metrics such as Accuracy, Precision, Recall, F1-Score, and R^2 (where R^2 denotes the coefficient of determination), optimizing hyperparameters via numerical methods.

3. Methodology

3.1. Data Source and Preprocessing

The primary dataset consists of 6 years (2020-2025) of surface water monitoring reports from EMB Region 6 covering 9 stations along the Jalaur River, categorized into Class A and Class C waters [1]. The data encompasses critical parameters including DO, BOD, Total Suspended Solids (TSS), pH, Temperature, Phosphates, Nitrates, and Fecal Coliform [1].

Data preprocessing will be executed using a Python pipeline utilizing the Pandas library. Missing values, common in sparse environmental datasets, will be estimated using KNN imputation, which relies on the Euclidean distance of available neighbor points [3]. Subsequently, a Random Forest (RF) algorithm will identify and filter non-linear outliers [3].

3.2. Feature Engineering and Target Generation

Not all water quality parameters contribute equally to the index. A comprehensive weight-based approach will be applied to prevent model overfitting [5]. This technique evaluates both the diversity of information (via Entropy Weighting) and the linear relationship between variables (via Pearson Correlation) [5].

The target variable for the classification algorithms will be the Water Quality Index (WQI). This will be computed using the Weighted Arithmetic WQI equation, derived from established water quality standards (e.g., WHO, DENR DAO 2016-08) [4]. The index equation is mathematically represented as:

$$WQI = \Sigma (q_i \times w_i) / \Sigma w_i$$

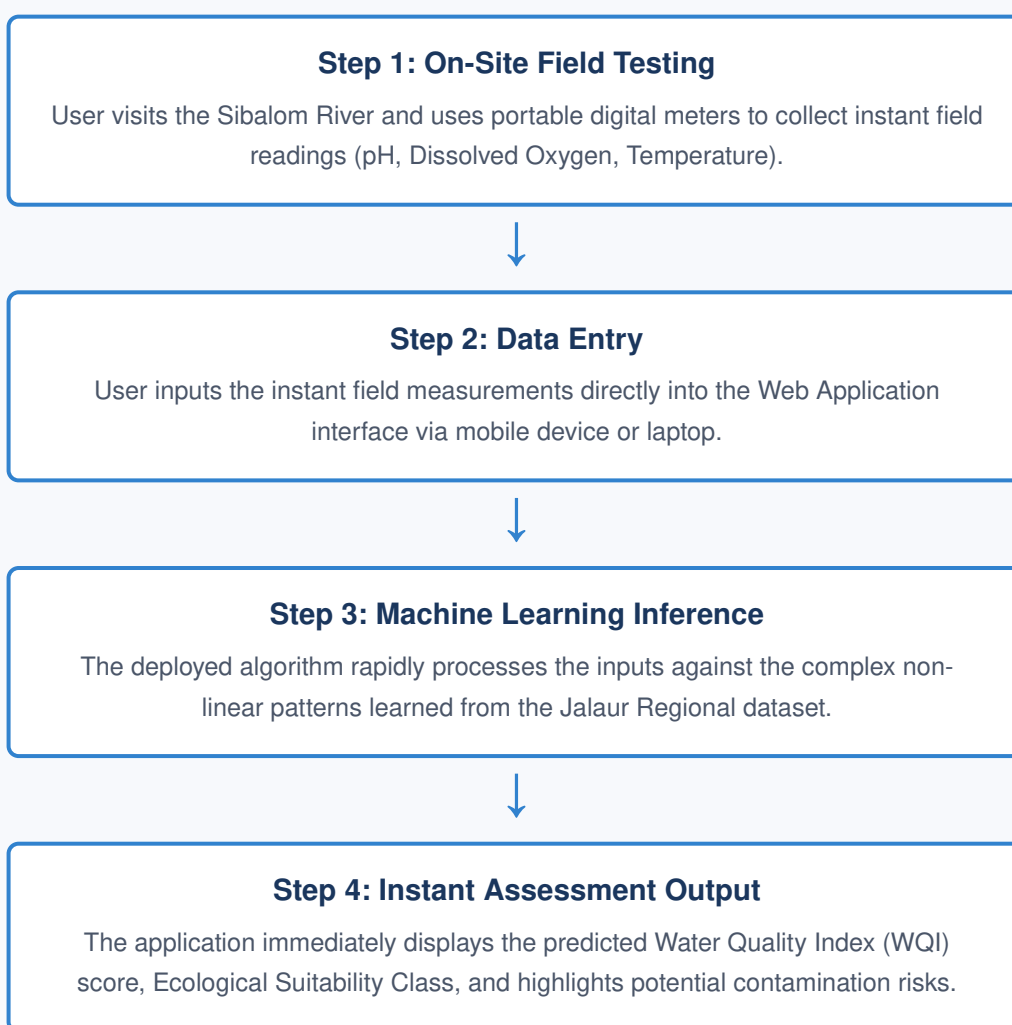
3.3. Validation vs. Deployment (The Role of the Laboratory)

A clear distinction is established between the research validation phase and the real-world application deployment. During the **validation phase**, physical water samples from the Sibalom River will be collected and sent for 5-day laboratory testing (e.g., BOD, Coliform) [2]. Concurrently, instant

measurements (pH, DO, Temperature) will be taken on-site [2]. The laboratory results serve strictly as the ground truth to prove the machine learning algorithm's predictive accuracy [3, 4]. Once the model's accuracy is statistically verified against these lab results, the laboratory requirement is entirely bypassed in the final product.

4. Web Application Deployment & User Workflow

The finalized machine learning model will be deployed as an interactive web application (e.g., utilizing Streamlit). This platform will enable environmental officers and local government units to instantly monitor water quality without the delays and costs associated with traditional laboratory processing. The end-user workflow is streamlined as follows:



5. Limitations of the Study

While the proposed machine learning framework offers a robust alternative to continuous laboratory testing, the study acknowledges several limitations:

- **Geographic Misalignment:** The model is trained predominantly on water quality data from the Jalaur River System Water Quality Management Area (JRS-WQMA) [1]. Unseen topological and land-use differences unique to the Sibalom River catchment might introduce variance that the model cannot fully anticipate without localized retraining.
- **Data Sparsity in Secondary Parameters:** Although primary field parameters like pH and Dissolved Oxygen are consistently logged, secondary parameters in the EMB Region 6 datasets (such as Fecal Coliform and Phosphates) often feature missing entries or non-detects [1], potentially constraining the algorithm's multi-variate correlations.
- **Hardware Dependency:** The accuracy of the deployed web application relies completely on the proper calibration and precision of the portable digital meters used by environmental officers in the field to gather the input features.

6. Expected Outcomes

The culmination of this study will be a validated, computationally optimized web application capable of predicting surface water quality classes with high accuracy. This tool will serve as a cost-effective, rapid-assessment alternative to traditional laboratory testing [3]. By successfully transferring regional data patterns from the Jalaur River to the Sibalom River, the project will provide local environmental authorities in Leon, Iloilo with a scalable, data-driven utility to monitor ecological health and enforce water management policies.

7. References

1. Environmental Management Bureau (EMB) Region 6. (2020-2025). *Summary Report of Water Quality Monitoring Data, Jalaur River System - Water Quality Management Area (JRS-WQMA)*.
2. Borlongan, I. G., Golez, N. V., & Lorque, F. (2010). Physico-Chemical Assessment of the Jalaur River System, Iloilo, Philippines. *Silliman Journal*, 51(1), 224-246.
3. Kularbphetpong, K., Raksuntorn, N., & Boonseng, C. (2025). Prediction of Water Quality Index (WQI) Using Machine Learning. *International Journal of Environmental Science and Development*, 16(1), 34-40.
4. Nair, J. P., & Vijaya, M. S. (2022). River Water Quality Prediction and index classification using Machine Learning. *Journal of Physics: Conference Series*, 2325, 012011.
5. Wang, X., Li, Y., Qiao, Q., Tavares, A., & Liang, Y. (2023). Water Quality Prediction Based on Machine Learning and Comprehensive Weighting Methods. *Entropy*, 25(8), 1186.